Qinhan Hou

Doctoral Researcher, University of Helsinki
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Date of CV: 25 October 2026

Degrees

Master of Science (MSc), Computer Science 01/09/2020–30/10/2022
Uppsala University, Uppsala, Sweden
Thesis: *A Pipeline of smFRET Data Processing Combined with NGS Technology*

Bachelor of Engineering (BEng), Software Engineering 01/09/2016–30/06/2020
Tianjin University, Tianjin, China
Thesis: *Decoding Music Genres Based on High Resolution Brain Activity Information*

Other Education and Expertise

LOGML Summer School 13/07/2026–17/07/2026
London, United Kingdom
Project: *An alternative perspective on Information Propagation in Graph Neural Networks*
Mentor: Giuseppe Alessio D'Inverno

Applied Deep Learning in Bioinformatics 11/08/2025–15/08/2025
Summer school, Copenhagen, Denmark

Project-based exchange, Data Science 01/04/2019–15/06/2019
ESILV – École Supérieure d'Ingénieurs Léonard de Vinci, Paris, France

Language Skills

- Mandarin Chinese: native
- English: full professional proficiency
- Italian: basic proficiency
- Finnish: beginner

Current Employment

Doctoral Researcher 01/09/2024–present
University of Helsinki, Finland

- Research focuses on graph neural networks and transformers, information propagation, positional and structural encodings, and continuous-depth models with adaptive graph topology.

Previous Work Experience

Junior Researcher 01/02/2023–31/08/2024
Swiss AI Lab IDSIA, Lugano, Switzerland

- Conducted machine learning research in the group of Prof. Jürgen Schmidhuber.
- Developed methods for silent speech interfaces using facial electromyography (EMG) signals.
- Contributed to self-supervised learning and EMG speech modelling, and supported teaching and master's student co-supervision.

Research Assistant 01/07/2021–30/10/2022
Deindl Lab, Uppsala University, Uppsala, Sweden

- Developed computational methods for single-molecule biophysics.
- Worked on image alignment, sequencing data processing, and machine learning for molecular time-series data.

Software Developer Intern 01/07/2019–31/08/2020
NetEase Inc., Beijing, China

- Developed and maintained components of a language translation software system.

Research Output

Journal articles

[1] J. Aguirre Rivera, G. Mao, A. Sabantsev, M. Panfilov, **Q. Hou**, et al. "Massively parallel analysis of single-molecule dynamics on next-generation sequencing chips." *Science*, 385, 892–898, 2024.

Conference and workshop papers

[2] **Q. Hou** and J. Tang. "Avoiding Feature Collapse in Graph ODEs via Hysteretic Topology Evolution." Accepted to *NeurIPS 2026, Main Track*. arXiv:2604.24293.

[3] **Q. Hou** and J. Tang. "Distance-Misaligned Training in Graph Transformers and Adaptive Graph-Aware Control." Extended abstract, *Graph Signal Processing Workshop (GSP)*, 2026. Poster presentation. arXiv:2604.22413.

[4] **Q. Hou**, Y. Zheng, X. Zhang, S. Luan, and J. Tang. "Exploring Heterophily in Graph-level Tasks." *NeurIPS 2025 Workshop on New Perspectives in Advancing Graph Machine Learning (NPGML)*, 2025. arXiv:2509.18893.

[5] **Q. Hou**, et al. "emg2vec: Self-supervised Pretraining in Electromyography-based Silent Speech Interfaces." *46th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, 2024. Oral presentation.

[6] Z. Ren, K. Scheck, **Q. Hou**, et al. "Diff-ETS: Learning a Diffusion Probabilistic Model for Electromyography-to-Speech Conversion." *46th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, 2024.

[7] K. Scheck, Z. Ren, T. Dombeck, J. Sonnert, S. van Gogh, **Q. Hou**, et al. "Cross-Speaker Training and Adaptation for Electromyography-to-Speech Conversion." *46th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, 2024.

[8] **Q. Hou** and G. Zhang. "Decoding Music Genres Based on High Resolution Brain Activity Information." *Asia-Pacific Signal and Information Processing Association Annual Summit and Conference (APSIPA ASC)*, 2020, pp. 1657–1665.

Research Supervision and Leadership Experience

Co-supervisor of Master's thesis 10/06/2025–present
University of Helsinki, Finland

- Co-supervised the Master's thesis of Xichun Zhang on heterophily in graph-level tasks and training dynamics of graph transformers.

Co-supervisor of Master's thesis 01/09/2023–30/04/2024
Università della Svizzera italiana (USI), Lugano, Switzerland

- Co-supervised the Master's thesis of Christian Squadrito on machine learning for EMG-based silent speech interfaces.

Teaching Experience

Lecturer 01/09/2025–present
University of Helsinki, Finland

- Delivered lectures in the course *Principles of Bioscience Omics*, focusing on machine learning methods for biomedical data.

Teaching Assistant 01/09/2023–31/01/2024
Università della Svizzera italiana (USI), Lugano, Switzerland

- Assisted with teaching, exercises, and student supervision for the Machine Learning course.

Awards and Honours

- Outstanding Student Award, Tianjin University, 2018–2019.
- Excellent Thesis Award, Tianjin University, 2020.
- International Student Representative, Uppsala University, 2022.

Other Key Academic Merits

Conference organisation

- Local Organising Committee, International Symposium on Bioinformatics Research and Applications (ISBRA), Helsinki, Finland, 2025.
- Local Organising Committee, 33rd International Conference on Artificial Neural Networks (ICANN), Lugano, Switzerland, 2023–2024.

Talks and presentations

- Poster presenter, *Distance-Misaligned Training in Graph Transformers and Adaptive Graph-Aware Control* (with Jing Tang), Graph Signal Processing Workshop (GSP), Madrid, Spain, 09/06/2026; workshop held 08/06/2026–10/06/2026.
- Oral presenter, IEEE EMBC 2024, Orlando, Florida, USA, 15/07/2024–19/07/2024.
- Oral presenter, APSIPA ASC 2020, Auckland, New Zealand, 07/12/2020–10/12/2020.

Peer review

- Reviewer for biomedical journals and computer science conferences, including NeurIPS 2025, ICML 2026, and NeurIPS 2026.
- Gold reviewer for ICML 2026.